

Probing Large Language Models for Scalar Adjective Lexical Semantics and Scalar Diversity Pragmatics

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Abstract

Scalar adjectives pertain to various domain scales and vary in intensity within each scale (e.g. *certain* is more intense than *likely* on the likelihood scale). Scalar implicatures arise from the consideration of alternative statements which could have been made. They can be triggered by scalar adjectives and require listeners to reason pragmatically about them. Some scalar adjectives are more likely to trigger scalar implicatures than others. This phenomenon is referred to as scalar diversity. In this study, we probe different families of Large Language Models such as GPT-4 for their knowledge of the lexical semantics of scalar adjectives and one specific aspect of their pragmatics, namely scalar diversity. We find that they encode rich lexical-semantic information about scalar adjectives. However, the rich lexical-semantic knowledge does not entail a good understanding of scalar diversity. We also compare current models of different sizes and complexities and find that larger models are not always better. Finally, we explain our probing results by leveraging linguistic intuitions and model training objectives.

Keywords: Scalar Adjective, Scalar Implicature, Lexical Semantics, Pragmatics

1. Introduction

Scalar adjectives (SAs) are words such as *likely*, *certain*, *warm*, and *scalding*. They describe different scales of properties. For instance, *warm* and *scalding* describe temperature, while *likely* and *certain* describe probabilities. Scalar adjectives can describe the same scale while differing in intensity. For example, *certain* is more intense than *likely* on the likelihood scale because it is used to make a logically stronger statement about a given situation.

Scalar implicatures (SIs) arise from the consideration of alternative statements that could have been made (Figure 1). They can be triggered by SAs. For instance, when a speaker utters ‘It is *likely* to rain’, a hearer may conclude that the speaker is *not certain* that it will rain. In particular, the hearer may reason that the speaker could have provided the logically stronger statement, ‘it is *certain* to rain’, but did not do so. Psycholinguistic studies indicate that some SAs are more likely to generate implicatures than others (Van Tiel et al., 2014; Gotzner et al., 2018; Ronai and Xiang, 2022). This phenomenon is referred to as *scalar diversity*. For instance, *likely* tends to indicate *not certain*, while *good* does not tend to indicate *not excellent*.

SI is a long-standing topic of research in pragmatics because it reveals fundamental aspects of human linguistic and cognitive capabilities in areas such as the Theory of Mind (Feng et al., 2021). Accordingly, SIs pose important challenges for the development of NLP models with human-like capabilities (Sap et al., 2022). SIs are also important in practice for downstream tasks that include Natural Language Inference (Williams et al., 2018), Sentiment Analysis (Socher et al., 2013), and indi-

rect Question Answering (de Marneffe et al., 2010). State-of-the-art large Language Models (LLM) such as GPT-4¹ have remarkable performance on many classic benchmarks. However, they have proved to be fragile in some semantic and pragmatic tasks that are easy for humans (Liu et al., 2023a; Lin et al., 2024). The goal of this paper is to probe this discrepancy with an in-depth investigation of SIs with SAs.

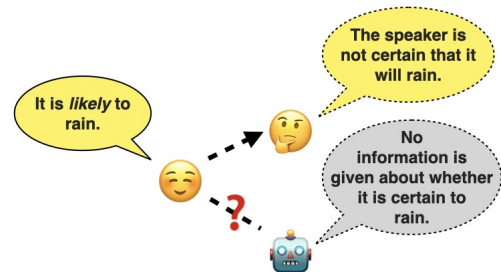


Figure 1: People often mean more than what they literally say. Humans can easily infer implied messages, while LLMs often fail to do so.

Recent research on LLMs’ understanding of SA has focused on: (i) probing the lexical semantics of SAs (Liu et al., 2023b) or (ii) their scalar diversity (Hu et al., 2022, 2023). In this paper, we investigate both (i) and (ii). We first utilize two probing methods to evaluate LLMs’ knowledge about SAs’ scale membership and intensity information. Using previously published datasets, we then assess whether LLMs show human-like scalar diversity in judgments about SA items. Finally, we compare LLMs’ pragmatic and lexical-semantic knowledge

¹<https://openai.com/research/gpt-4>

and explain our observations based on linguistic theory.

Our paper includes three main findings. First, LLMs generally encode rich lexical-semantic information about SAs (§3). Second, LLMs have unsatisfying performance in capturing scalar diversity despite encoding rich lexical-semantic information about SAs (§4). Third, the size of the LLMs does not correlate with how well they perform on our tasks (§3 and §4): While the increase in model size is sometimes claimed to invariably improve performance (the so-called "scaling law"), in our study, some large models do worse than smaller models that have different architectures or training objectives.²

2. Related works

2.1. Scalar Adjective Lexical Semantics

Since the seminal work by Kamp (1975), SAs have received a lot of attention in formal semantics and the philosophy of language. The same cannot be said about NLP research, where SAs have only recently garnered attention. Some of this research has been concerned with the lexical semantics of SAs, particularly focusing on their scale membership and their intensities (de Melo and Bansal, 2013; Kim and de Marneffe, 2013; Shivade et al., 2015; Wilkinson and Tim, 2016; Cocos et al., 2018). One feature of this research is the use of static rather than contextualized word embeddings from LLMs. This is important because Garí Soler and Apidianaki (2020) have found that BERT-base (Devlin et al., 2019) contextualized word embeddings encode richer information about scalar intensities in vector space. Nevertheless, Liu et al. (2023b) report that many state-of-the-art models, even after fine-tuning on MNLI (Williams et al., 2018), have unsatisfying adjective degree estimations in textual inference.

2.2. Scalar Implicature and Scalar Diversity Pragmatics

In a study of SIs, Schuster et al. (2020) have found that LSTM (Hochreiter and Schmidhuber, 1997) can be trained to infer *some-not all* implicatures (e.g. 'Some student cheated' implicates that not all students cheated). Jeretic et al. (2020) report that BERT fine-tuned on MNLI nearly always predicts that the determiner *some* entails *not all*, but that SAs are treated as synonyms regardless of their intensity. This means that the model may lack relevant pragmatic knowledge about adjectives. In contrast, two recent studies (Hu et al.,

2022, 2023) achieved more favorable results with surprisal measures derived from GPT2. String-based surprisal, which considers the surprise level of a given strong word to appear as an alternative to a weak word out of all possible strings in a context, is estimated by the likelihood of a strong word to appear in its position by a GPT-2 model (Radford et al., 2019). Concept-based surprisal treats an alternative as a member of a string set with similar concepts. The surprisal rate for the alternative is considered as an average over that of all strings in such a set. Using these two measures, they found that scalar diversity correlates with string-based and concept-based surprisal. Ruis et al. (2024) tested different LLMs on general conversational implicature understanding. They found that GPT-4 is the best-performing model with 30-shot chain-of-thought prompting (Wei et al., 2022), achieving 88.66% in generalized implicature calculation. This is the category that our adjective-triggered SIs fall under.

2.3. Direct and Indirect Probing

Direct and indirect probing methodologies are widely used to understand LLMs' knowledge. Direct probing is used to analyze the hidden representations encoded in LLMs (Garí Soler and Apidianaki, 2020). This is only possible when the LLM is open-source. Indirect probing is used to analyze the representations of closed-source models. This involves testing performance on various semantic tasks. Typical tasks use textual prompts to assess the prompted answers (Ettinger, 2020). Petroni et al. (2019) argue that indirect probing only reveals the lower bound of model capabilities. Follow-up works have provided various methods to increase the bound (Zhong et al., 2021; White et al., 2023).

We build on these results as follows. Since direct probing provides a good estimate of a model's competence, but closed-source LLMs only allow for indirect probing, we use both methods for some open-source models. We then use the results on these models to ground the results of indirect probing for other models. We make the following assumption: if the two probing methods show similar trends regarding model performance for the open-source models, we deem our indirect probing methods as valid in comparing the relative capabilities of different closed-source models.

3. Probing Lexical Semantics

Recent research on the lexical semantics of SAs have mostly focused on deriving their intensity information from open-source models (Garí Soler and Apidianaki, 2020; Liu et al., 2023b). In this section, we describe how we probe three model families, for

²Code is in https://github.com/fangru-lin/llm_scalar_adj.

two aspects of SA lexical semantics: (i) scale membership (e.g. *likely* and *certain* are on the likelihood scale), and (ii) adjective intensity (e.g. *certain* is more intense than *likely* within the same scale).

We assess three model families, containing eight models in total, to understand how different model architectures, training objectives, and sizes affect lexical-semantic knowledge. All models except GPT-4 are accessed via huggingface library (Wolf et al., 2020), and GPT-4 is accessed via OpenAI API.³

Encoder Models BERT-base/large(b/l) (110M/340M) (Devlin et al., 2019), RoBERTa-base/large(b/l) (123M/354M) (Liu et al., 2019)

Decoder Models Falcon-7B-instruct (Falcon) (Almazrouei et al., 2023), GPT-4⁴

Encoder-decoder Models Flan-T5-xl/xxl (3.5B/11B) (Chung et al., 2022)

3.1. Datasets for Lexical-Semantic Probing

3.1.1. Scalar Adjective Datasets

Following Garí Soler and Apidianaki (2020), we use three SA datasets: DEMELO (**DM**) (de Melo and Bansal, 2013), CROWD (**CD**) (Cocos et al., 2018), and WILKINSON (**WK**) (Wilkinson and Tim, 2016) which contain 185 half-scales in total (Table 1). For example, the positive side of the quality scale ranges from *good* to *awesome*. Each contains different categories of SAs. Some contain adjectives describing physical appearance while others do not.

	DM	CD	WK
Half-scales	87	77	21
Distinct adjective pairs	548	330	61

Table 1: Overview of half-scale counts and distinct adjective pairs in SA datasets.

DM Half-scales are first collected from WordNet 3.0 (Gross, Derek and Miller, Katherine J, 1990) and then annotated for intensity by two native English speakers.

CD Adjectives are first collected from the Paraphrase Data Base (Ganitkevitch et al., 2013; Pavlick et al., 2015) and then annotated by crowd workers. Annotators are asked to identify whether given adjectives are on the same scale for multiple rounds, and then annotate the intensity of adjectives.

WK Adjectives are collected via crowdsourcing. Crowd workers are first presented with prompt words that belong to a full scale and asked to list

other semantically related adjectives. Scales are, then, cleaned automatically based on workers’ competence and annotated manually for adjective intensities. This study uses partitioned **WK** with half scales in Cocos et al. (2018), which are derived from the Paraphrase Data Base (Ganitkevitch et al., 2013; Pavlick et al., 2015).

3.1.2. Context Sentence Datasets

To compute contextualized word embeddings for adjectives, we use the context sentence dataset **ukWac** from Garí Soler and Apidianaki (2020). **ukWac** provides 10 context sentences for each half-scale s in **DM**, **CD**, and **WK**. All adjectives on s share the same context set of 10 different sentences. Each candidate sentence contains an adjective on s and allows lexical substitution of its scale-mates. The acceptability after substitution is ensured by computing their acceptability scores using context2vec (Melamud et al., 2016). An example is given below, where the italic word is an adjective on s , and appears in the original context sentence. Other scale alternatives to it are inside the brackets.

They are extremely catchy
and are not only great to
listen to, but they are
also *thrilling* (interest-
ing/moving/exciting) to sing.

3.2. Probing Scale Membership

In this subsection, we assess whether LLMs encode the notion of scale membership. Ideally, we would expect models to be able to assign adjectives to their corresponding scales in context (e.g. *warm* belongs to the temperature scale).

3.2.1. Scale Membership Direct Probing Method

We directly probe scale membership by getting contextualized representations of SAs and half-scales and calculating their similarities. We retrieve word embeddings using **ukWac** by sampling different sentences for each adjective a on a half-scale s to obtain its contextualized representation $\vec{a}$. We derive a scale vector $\vec{s}$ for each s by adding the representations of the weakest and strongest adjectives on s . For instance, for the scale *adequate-fine-fitting-good*, where *adequate* is the least intense and *good* is the most intense adjective, $\vec{s}$ is $\vec{adequate} + \vec{good}$. Then, the cosine similarity $\cos(\vec{a}, \vec{s})$ between each adjective a in a SA dataset D_a and all other $\vec{s}$ in the dataset is computed. The scales are ranked by $\cos(\vec{a}, \vec{s})$ for each adjective; the scale with the highest $\cos(\vec{a}, \vec{s})$ is considered to be the most likely scale for a to belongs to, and so

³See appendix 11.6 for more implementation details.

⁴We do not use LLaMA (Touvron et al., 2023) due to our institutional requirement. We use gpt-4-0613 for GPT-4.